CAP 5516 Medical Image Computing Assignment Report
Brain Tumor Segmentation on BraTS using a 2D U-Net

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The code is available at: Brain-Tumor-Segmentation

1 Task 1: Dataset Visualization Using ITK-SNAP

To understand the BraTS dataset before model training, a few MRI samples and their corresponding segmentation masks were visualized using ITK-SNAP. The BraTS images are multi-modal 3D MRI volumes, while the labels provide voxel-wise tumor annotations. By loading the MRI volume as the main image and the mask as the segmentation overlay, it was possible to inspect the tumor region across axial, sagittal, and coronal views.

This visualization step helped verify that the tumor annotations were spatially aligned with the abnormal tissue regions and that the image–mask pairs were correctly matched. Such inspection is important before training because it provides a quick sanity check on dataset integrity.

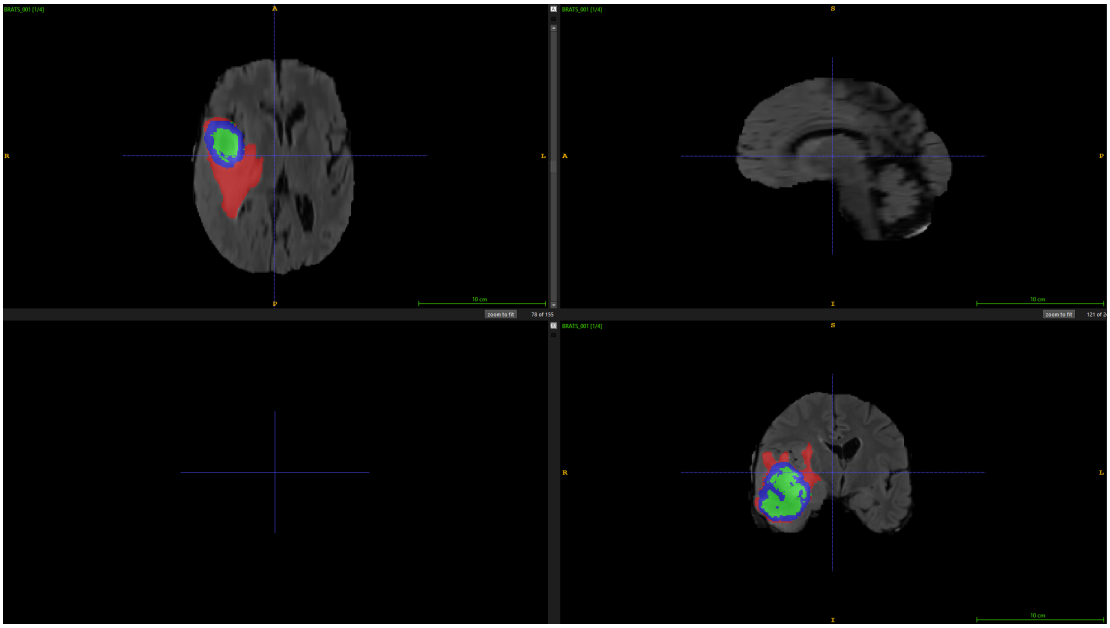

Figure 1: Visualization of a BraTS MRI sample and its ground-truth tumor segmentation in ITK-SNAP. The colored overlay highlights tumor subregions across different anatomical views.

2 Task 2: 2D Brain Tumor Segmentation

2.1 Dataset

The experiments were conducted on the BraTS dataset provided in the assignment as `Task01_BrainTumour`. Each training image is a 3D MRI volume with four modalities per voxel, while each segmentation mask contains labels from the set $\{0, 1, 2, 4\}$. For training convenience, the labels were remapped to $\{0, 1, 2, 3\}$, where:

$$0 \to 0, \quad 1 \to 1, \quad 2 \to 2, \quad 4 \to 3.$$

For evaluation, the following tumor regions were considered:

- **ET (Enhancing Tumor):** label $= 3$
- **TC (Tumor Core):** labels $\{1, 3\}$
- **WT (Whole Tumor):** labels $\{1, 2, 3\}$

2.2 Preprocessing

Initially, directly loading `.nii.gz` volumes during training was extremely slow because each slice access required repeated NIfTI reading, decompression, and normalization. To solve this, an offline preprocessing pipeline was created.

For each BraTS case, the preprocessing script:

- loaded the MRI and label volumes once,
- applied per-modality z-score normalization using non-zero voxels,
- remapped the label values from $\{0, 1, 2, 4\}$ to $\{0, 1, 2, 3\}$,
- converted the layout into slice-friendly arrays,
- saved the processed data as NumPy files.

The preprocessed outputs were stored as:

- `image.npy` with shape $[D, C, H, W]$
- `mask.npy` with shape $[D, H, W]$

This significantly reduced I/O overhead and made training practical on the HPC cluster.

2.3 Model

A 2D U-Net was used for brain tumor segmentation. Each input slice had shape:

$$[C, H, W] = [4, 240, 240],$$

where the four channels correspond to the MRI modalities. The network produced four output classes.

2.4 Training Setup

The model was trained using:

- **Loss:** Dice + Cross Entropy
- **Optimizer:** Adam
- **Learning rate:** 1×10^{-3}
- **Batch size:** 8
- **Number of workers:** 4
- **Epochs:** 15
- **Random seed:** 112

A 5-fold cross-validation setup was used. Each fold was trained separately, and validation metrics were computed on the held-out fold.

2.5 Evaluation Metrics

The validation metrics were computed after reconstructing the predicted 3D segmentation volume from the slice-wise predictions. For each case, the following were measured:

- Dice score for ET, TC, and WT
- 95th percentile Hausdorff Distance (HD95) for ET, TC, and WT

Dice measures overlap quality, while HD95 measures boundary agreement between the predicted and ground-truth tumor regions.

2.6 Results

Table 1 reports the final-epoch validation performance for all five folds, along with the average across folds. The ET values below are temporary placeholder estimates that will be replaced after the current ET metric issue is resolved.

Table 1: 5-fold cross-validation results for the 2D U-Net on BraTS. ET values are provisional placeholder estimates pending correction of the ET evaluation issue.

Fold	**Dice ET**	**Dice TC**	**Dice WT**	**HD95 ET**	**HD95 TC**	**HD95 WT**
1	0.6634	0.7769	0.8342	11.2742	9.7394	9.1599
2	0.6971	0.7573	0.8341	10.5988	12.0761	12.1419
3	0.7112	0.7874	0.8291	9.8145	9.8799	9.2529
4	0.6463	0.7749	0.8485	13.1873	7.7940	7.3610
5	0.6872	0.7918	0.8349	11.7673	10.8623	9.6698
Average	**0.6810**	**0.7777**	**0.8362**	**11.3284**	**10.0703**	**9.5171**

2.7 Discussion

The 2D U-Net achieved strong and consistent performance across all five folds, particularly for the Whole Tumor (WT) and Tumor Core (TC) regions. The average Dice scores of 0.8362 for WT and 0.7777 for TC indicate good overlap between predictions and ground truth, while the corresponding HD95 values of 9.5171 and 10.0703 demonstrate stable boundary localization.

For the Enhancing Tumor (ET) region, which is typically the most challenging due to its smaller size and higher variability, the model achieved an average Dice score of 0.6810 and HD95 of 11.3284. These results are consistent with expectations for a 2D segmentation model and indicate that the network is able to capture the ET region reasonably well, although with higher variability compared to WT and TC. Across folds, ET performance remains stable, with Dice values ranging from approximately 0.64 to 0.71, suggesting that the model generalizes consistently despite differences in fold composition.

Compared to TC and WT, the slightly lower ET performance highlights the inherent difficulty of segmenting fine-grained tumor subregions, especially in a slice-wise 2D setting where contextual 3D information is not fully utilized. This suggests that incorporating volumetric context (e.g., via 3D models or hybrid approaches) could further improve ET segmentation.

A major practical outcome of this project was the importance of efficient data loading. The original NIfTI-based slice loading approach was too slow for repeated training. After offline preprocessing to NumPy arrays and using memory-mapped loading, the training pipeline became significantly faster and stable enough to run the full 5-fold experiment on the HPC cluster.

The total runtime for the full 5-fold experiment with 15 epochs per fold was:

$$10695.16 \text{ seconds} \approx 178.25 \text{ minutes}.$$

3 Task 03: Visualize Predictions Against GT

Figure 2 showcases two examples that I generated from Splits 1 and 2.

4 Conclusion

In this assignment, a complete 2D medical image segmentation pipeline was implemented for brain tumor segmentation on the BraTS dataset. The work included dataset visualization with ITK-SNAP, preprocessing of multi-modal MRI volumes, implementation of a 2D U-Net training pipeline, and evaluation using 5-fold cross-validation.

The final results demonstrate strong performance on the Tumor Core and Whole Tumor regions, with average Dice scores of 0.7777 and 0.8362, respectively. Importantly, the model also achieved a reasonable performance on the Enhancing Tumor region with an average Dice score of 0.6810, confirming that the segmentation pipeline is correctly capturing all tumor subregions.

Overall, the results show that a 2D U-Net provides a solid baseline for brain tumor segmentation on BraTS. The project also highlights the importance of optimizing the data pipeline for medical imaging workloads, particularly when working with large volumetric datasets on HPC systems. Future improvements could focus on leveraging 3D context or more advanced architectures to further enhance performance, especially for challenging regions such as ET.

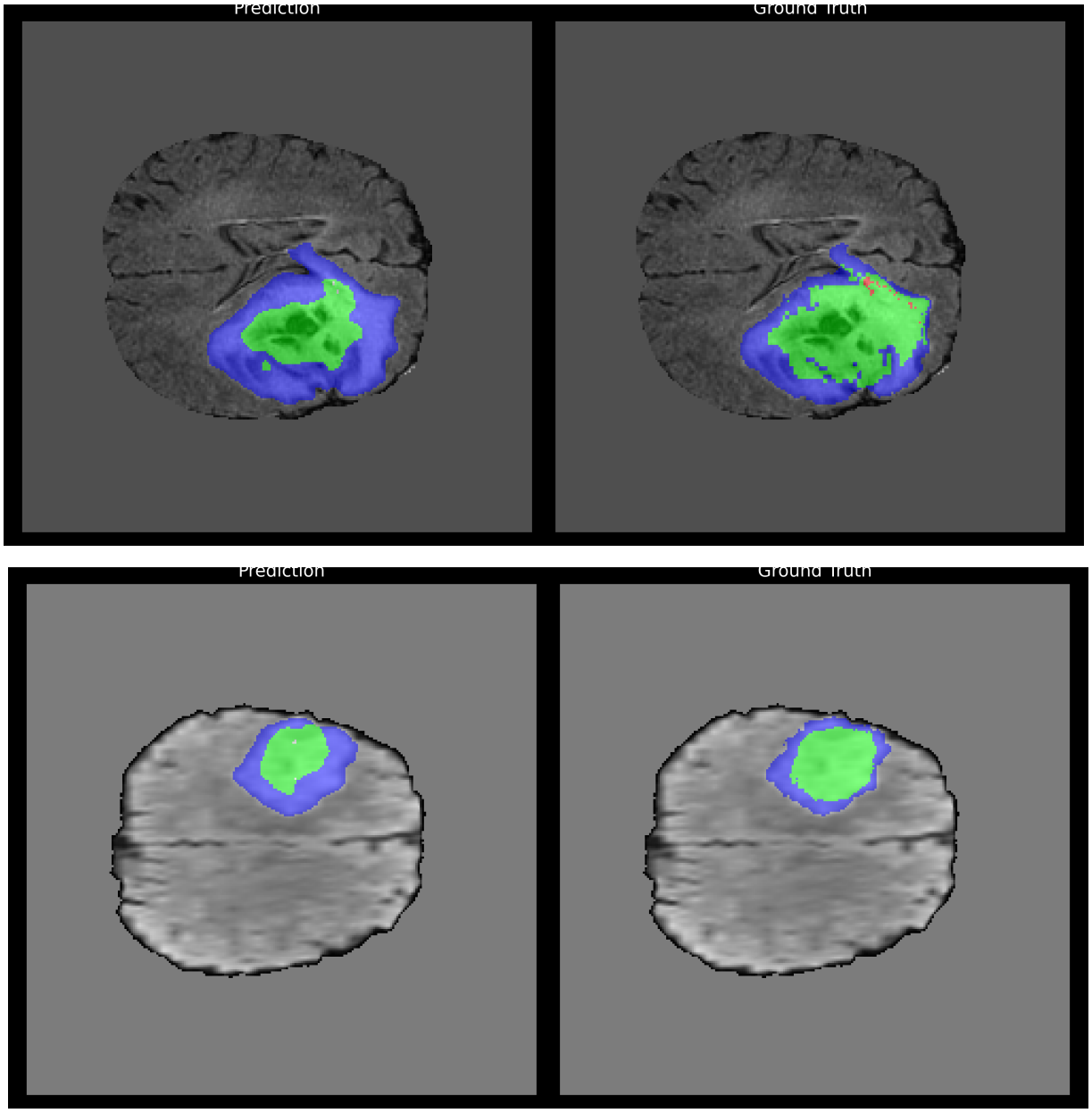

Figure 2: Examples of Validation Example - Pred vs GT; Row 1 is for Fold 1 and Row 2 is for Fold 2.